

Time series

$$X_t, t \in \{1, \dots, n\}$$



Estimation of $\hat{\Gamma}$ and $\hat{\Delta}$

$$\hat{\Gamma} = \sum_{k=-M}^M \lambda(k/M) k^2 \hat{R}_{\text{IF}}(k) \frac{(\omega * \omega)''(0)}{2(\omega * \omega)(0)}$$

$$\hat{\Delta} = 2\hat{\sigma}_{\infty}^4 \int_{-1}^1 \frac{(\omega * \omega)^2(x)}{(\omega * \omega)^2(0)} dx$$

where, $\hat{\sigma}_{\infty}^2 = \sum_{k=-M}^M \lambda(k/M) \hat{R}_{\text{IF}}(k)$, and M is found using Politis (2003), as in the cPW method.



Definition of $\hat{R}_{\text{IF}}(k)$

$$\hat{R}_{\text{IF}}(k) = n^{-1} \sum_{i=1}^{n-|k|} (X_i - \bar{X}_n) (X_{i+|k|} - \bar{X}_n)$$

where, $k \in \{-M, \dots, M\}$

Definition of $\lambda(t)$

$$\lambda(t) = \begin{cases} 1, & |t| \in [0, 1/2] \\ 2(1 - |t|), & |t| \in [1/2, 1] \\ 0, & \text{Otherwise} \end{cases}$$

Definition of $w_c(t)$

$$w = w_c^{\text{TRAP}}(t) = \begin{cases} t/c, & |t| \in [0, c] \\ 1, & |t| \in [c, 1 - c] \\ (1 - t)/c, & |t| \in [1 - c, 1] \end{cases}$$

Calculate Optimal Block Length

$$\hat{\ell}_{TBB}^{\text{opt}} = \hat{\ell}_{ETBB}^{\text{opt}} = \left[\frac{4\hat{\Gamma}^2}{\hat{\Delta}} \right]^{\frac{1}{5}} n^{\frac{1}{5}}$$